

How to read the data from the client :

In order to read the data from the client we can use the following techniques :

- By using java.util.Scanner class [Available from JDK 1.5V]
- By using java.lang.IO class [Available from JDK 25V]

Reading the data through IO class :

- IO is a predefined utility class available from JDK 25V and used to perform input and output operation.
- It provides various **static methods** to perform input & output operation.

a) public static String readln() :

- It is a predefined static method of IO class.
- Used to read a **complete line** and **return the data in String format**.
- Using this method, we can provide a prompt to the user for asking the input data.

b) public static String readln(String prompt) :

- It is a predefined static method of IO class.
- Used to read a **complete line** and **return the data in String format**.
- Using this method, we can provide a prompt to the user for asking the input data.

WAP to read your name from the keyboard using readln() method of IO class :

```
ReadName.java
-----
void main()
{
    IO.print("Enter your Name :");
    String name = IO.readln();
    IO.println("Your Name is :"+name);
}
```

WAP to read your name from keyboard using readln() method with prompt (user input message) :

```
ReadName.java
-----
void main()
{
    String name = IO.readln("Enter your Name :");
    IO.println("Your Name is :"+name);
}
```

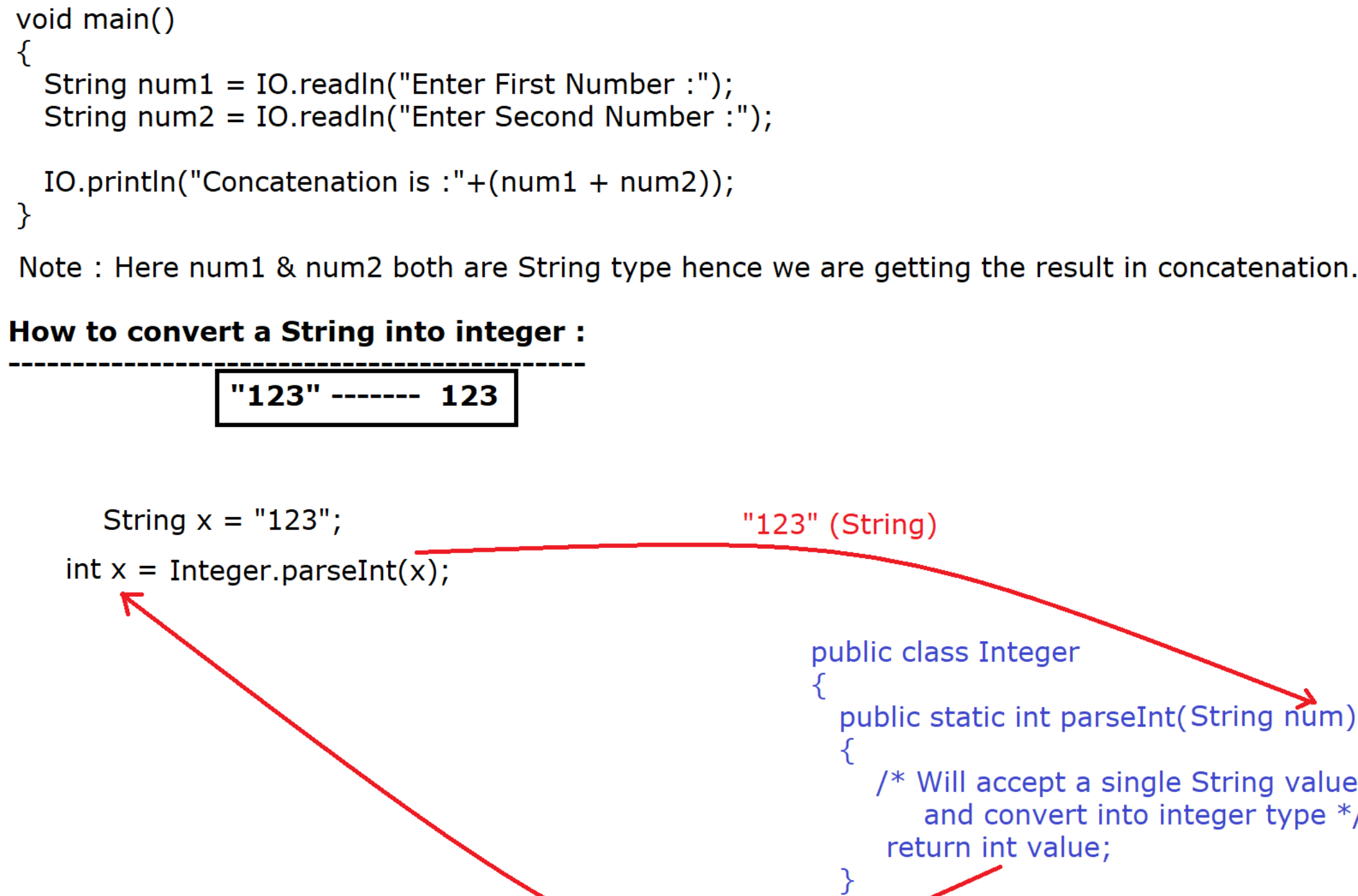
WAP to add two numbers by using user input :

```
void main()
{
    String num1 = IO.readln("Enter First Number :");
    String num2 = IO.readln("Enter Second Number :");
    IO.println("Concatenation is :"+(num1 + num2));
}
```

Note : Here num1 & num2 both are String type hence we are getting the result in concatenation.

How to convert a String into integer :

"123" ----- 123



In order to convert the String value into integer, we can use a predefined class called **Integer** available in **java.lang** package.

This Integer class has provided predefined static method called **parseInt(String num)** through which we can convert a String value into integer type.

After conversion It will return int value because parseInt(String num) method, return type is integer.

```
Addition.java
-----
void main()
{
    String x = IO.readln("Enter first number :");
    String y = IO.readln("Enter second number :");

    //Converting String into integer
    int num1 = Integer.parseInt(x);
    int num2 = Integer.parseInt(y);

    IO.println("Addition is :"+(num1 + num2));
}
```

```
FindingCube.java :
-----
void main()
{
    int num = Integer.parseInt(IO.readln("Enter a number :"));
    IO.println("Cube of "+num+" is :"+(num*num*num));
}
```

```
Rectangle.java
-----
void main()
{
    int length = Integer.parseInt(IO.readln("Enter the length :"));
    int breadth = Integer.parseInt(IO.readln("Enter the breadth :"));

    double areaOfRect = length * breadth;

    IO.println("Area of Rectangle is :"+areaOfRect);
}
```

How to convert a String into float :

In order to convert a String into float type, we can use Float.parseFloat()

How to convert a String into double :

In order to convert a String into double type, we can use Double.parseDouble()

FindingAreaOfCircle :

```
void main()
{
    final double PI = 3.14;
    double radius = Double.parseDouble(IO.readln("Enter the radius :"));
    double areaOfCircle = PI * radius * radius;
    System.out.printf("Area of Circle is :%.2f",areaOfCircle);
    IO.println();
}
```

WAP to read your gender (M/F) from the keyboard :

String class has provided a non-static method called **charAt(int index)**, using this we can retrieve a character based on the given index. The return type of this method is char.

```
public char charAt(int indexPosition)
```

```
ReadCharacter.java
-----
void main()
{
    String str = "Java";
    char ch = str.charAt(0);
    IO.println(ch); //J
}
```

```
ReadGender.java
-----
//Reading the gender from the keyboard
void main()
{
    char gender = IO.readln("Enter your Gender :").charAt(0);
    IO.println("Yout gender is :"+gender);
}
```

```
ReadBoolean.java
-----
//Reading the boolean value from the keyboard
void main()
{
    String booleanValue = IO.readln("Enter boolean value [true /false] :");
    boolean isValid = Boolean.parseBoolean(booleanValue);
    IO.println("You have entered :"+isValid);
}
```

```
ReadAge.java
-----
//Reading the age from the keyboard
void main()
{
    int age = Integer.parseInt(IO.readln("Enter your Age :"));

    if(age > 18)
    {
        IO.println("Eligible for a Movie");
    }
    else
    {
        IO.println("Not Eligible for a Movie");
    }
}
```

Read gender in String and provide a welcome message :

In order to compare two Strings we can use equals(Object obj) method of String class. Return type of this method is boolean.

```
public boolean equals(Object obj)

//Reading gender in String
void main()
{
    String gender = IO.readln("Enter your gender [Male/Female] :");

    if(gender.equals("Male"))
    {
        IO.println("Welcome to this application Sir!!!");
    }
    else
    {
        IO.println("Welcome to this application Madam!!!");
    }
}
```

Boolean Literal :

- It has two states true OR false.
- It is used to represent flag value. [on OR off]
- It has only data type i.e. boolean data type which accepts 1 bit of information OR depends upon JVM implementation.

Example :

```
boolean isEmpty = false;
boolean isValid = true;
```

Unlike C and C++, we cannot assign 0 OR 1 to the boolean variable because in java 0 and 1 represent integral literal.

```
boolean x = 0; [Valid in C and C++ but invalid in java]
```

A string value, we cannot assign to boolean variable.

```
boolean x = "true";
```